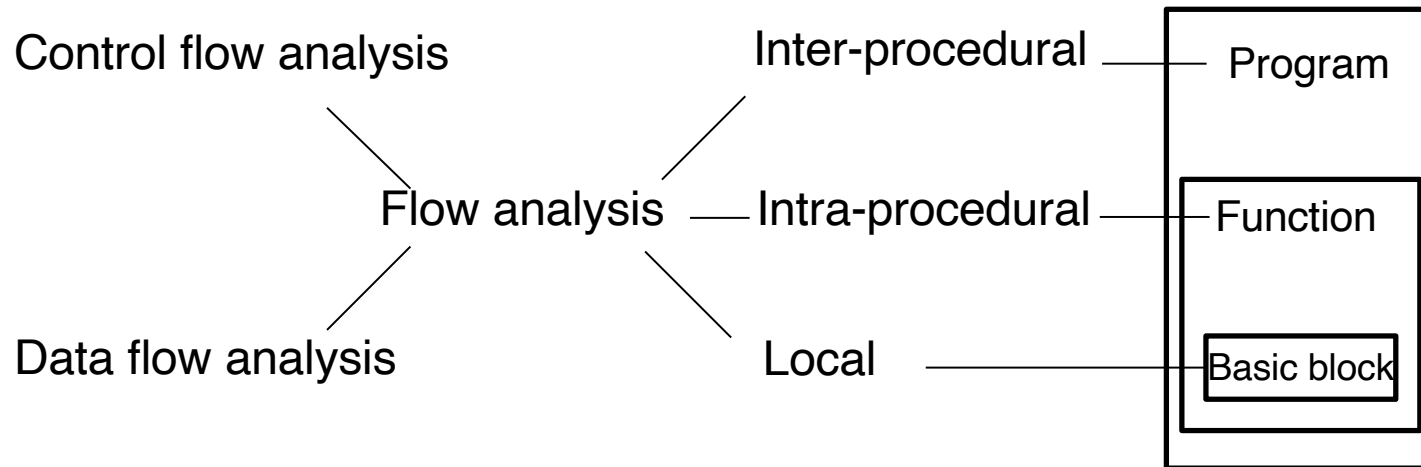


Control Flow Graph

Program Flow Analysis



- ▶ Control flow analysis: determine control structure of a program and build control flow graphs (**CFGs**)
- ▶ Data flow analysis: determine the flow of data values and build data flow graphs (**DFGs**)

Basic Blocks

- ▶ **Basic block:** a sequence of consecutive intermediate language statements in which flow of control can only enter at the beginning and leave at the end
 - Only the last statement of a basic block can be a branch statement and only the first statement of a basic block can be a target of a branch

Partitioning a Program into Basic Blocks

- ▶ Each basic block begins with a leader statement
- ▶ Identify leader statements (i.e., the first statements of basic blocks) by using the following rules:
 - (i) The **first statement** in the program is a leader
 - (ii) Any statement that is the **target of a branch statement** is a leader (for most intermediate languages these are statements with an associated label)
 - (iii) Any statement that **immediately follows a branch or return** statement is a leader

Example: Forming the Basic Blocks

(1) initialize(...)
(2) if (theta > curr) goto (7)
(3) T = X + (Y >> i);
(4) Y = -(X >> i) + Y;
(5) curr = curr - ctab[i];
(6) goto (10)
(7) T = X - (Y >> i);
(8) Y = (X >> i) + Y;
(9) curr = curr + ctab[i];
(10) X = T;
(11) i++
(12) if (i < STEPS) goto (2)
(13) return X, Y

Leader statement is:

- (1) the first in the program
- (2) any that is the target of a branch
- (3) any that immediately follows a branch

Basic Blocks:

B1 (1) initialize(...)

B2 (2) if (theta > curr) goto (7)

B3 (3) T = X + (Y >> i);
(4) Y = -(X >> i) + Y;
(5) curr = curr - ctab[i];
(6) goto (10)

B4 (7) T = X - (Y >> i);
(8) Y = (X >> i) + Y;
(9) curr = curr + ctab[i];

B5 (10) X = T;
(11) i++
(12) if (i < STEPS) goto (2)

B6 (13) return X, Y

Control Flow Graph (CFG)

- ▶ A **control flow graph** (CFG), or simply a flow graph, is a directed graph in which:
 - (i) the nodes are basic blocks; and
 - (ii) the edges are induced from the possible flow of the program
- ▶ The basic block whose leader is the first intermediate language statement is called the **entry node**
- ▶ In a CFG we assume no information about data values
 - an edge in the CFG means that the program **may** take that path

Example: Control Flow Graph Formation

B1 (1) initialize(...)

B2 (2) if (theta > curr) goto (7)

B3 (3) $T = X + (Y \gg i);$
(4) $Y = -(X \gg i) + Y;$
(5) $\text{curr} = \text{curr} - \text{ctab}[i];$
(6) goto (10)

B4 (7) $T = X - (Y \gg i);$
(8) $Y = (X \gg i) + Y;$
(9) $\text{curr} = \text{curr} + \text{ctab}[i];$

B5 (10) $X = T;$
(11) $i++$
(12) if ($i < \text{STEPS}$) goto (2)

B6 (13) return X, Y

